

# Development of a Multiplayer Roleplay Server for GTA V Using FiveM and Qbox Framework

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## ABSTRACT

Bangladesh lacks advanced, locally hosted multiplayer roleplay servers. This project delivers a production-ready GTA V roleplay server on FiveM using the Qbox framework. A modular, event-driven, three-tier architecture integrates 50 resources for jobs, economy, phone, and voice services, backed by optimized MySQL. Server-authoritative validation and asynchronous data operations improve stability, security, and responsiveness.

## INTRODUCTION AND PROBLEM

Existing public and legacy servers often suffer from:

- poor system integration and inconsistent play
- performance degradation under player load
- security exploits and item or money duplication
- limited scalability and weak local relevance

The project provides a secure, culturally localized alternative with Bangladeshi maps, identity cards, banking, hospital branding, and community-focused roleplay.

## KEY FEATURES

13 job systems: police, EMS, and civilian careers with progression.  
Full economy: banking, ATMs, vehicles, property, garage, and inventory.  
NPWD phone: wallet, garage, maps, messaging, and services.  
3D voice and radio: proximity communication and job channels.  
Security: server-side validation, rate limiting, and mutex locks.

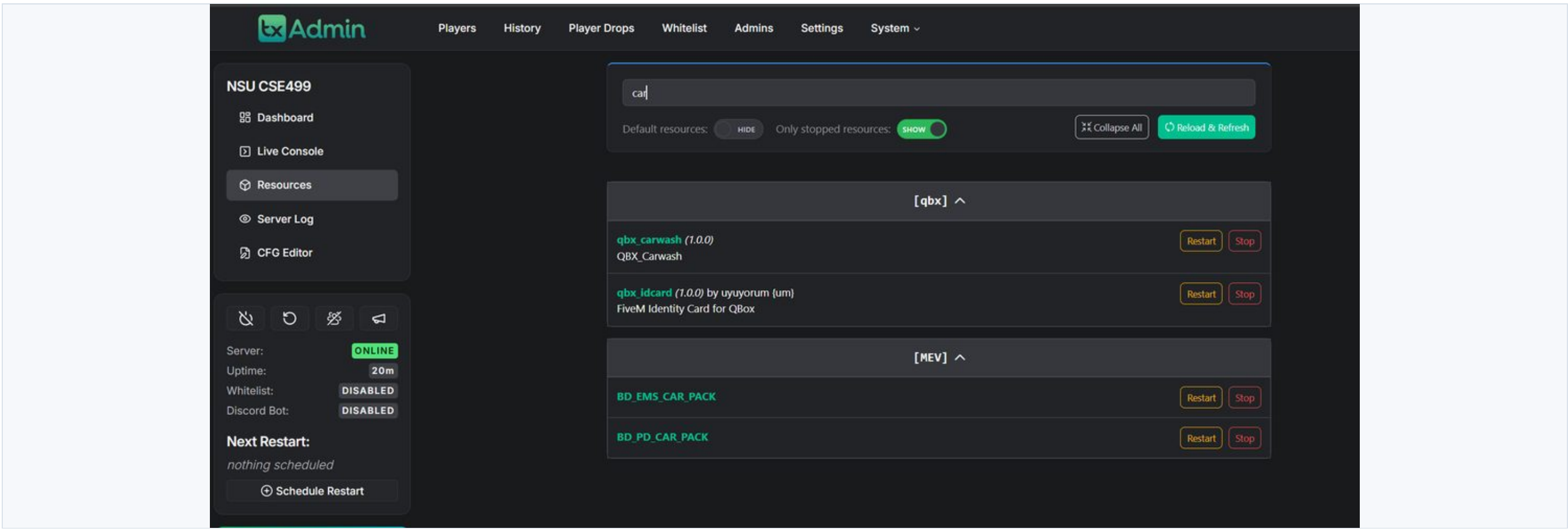
## PROJECT OBJECTIVES

1. Integrate a stable Qbox-based multiplayer platform.
2. Optimize database access to below 50 ms query latency.
3. Validate gameplay, economy, communication, and security under realistic load.
4. Establish a scalable foundation for Bangladesh's gaming and software community.

## SYSTEM INTERFACES

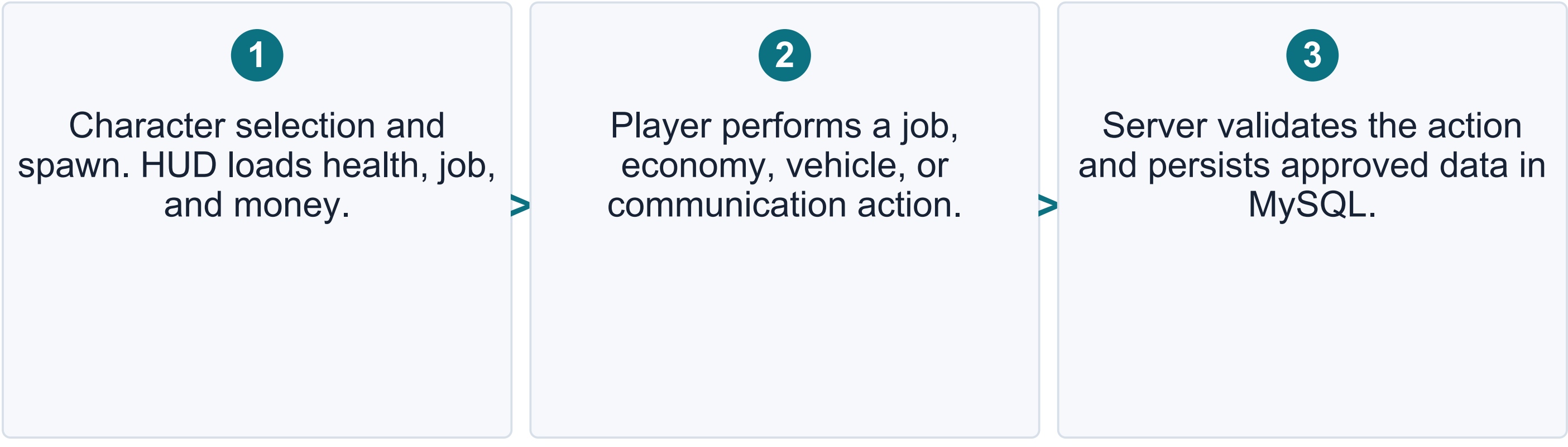
## METHODOLOGY

Architecture: three-tier client, application, and data design with 50 integrated Qbox resources.  
Tools: Lua 5.4, Qbox, oxmysql, ox\_lib, ox\_inventory, NPWD, pma-voice, Git, and txAdmin.  
Process: plan, core setup, feature integration, security and stress testing, deployment, and documentation.

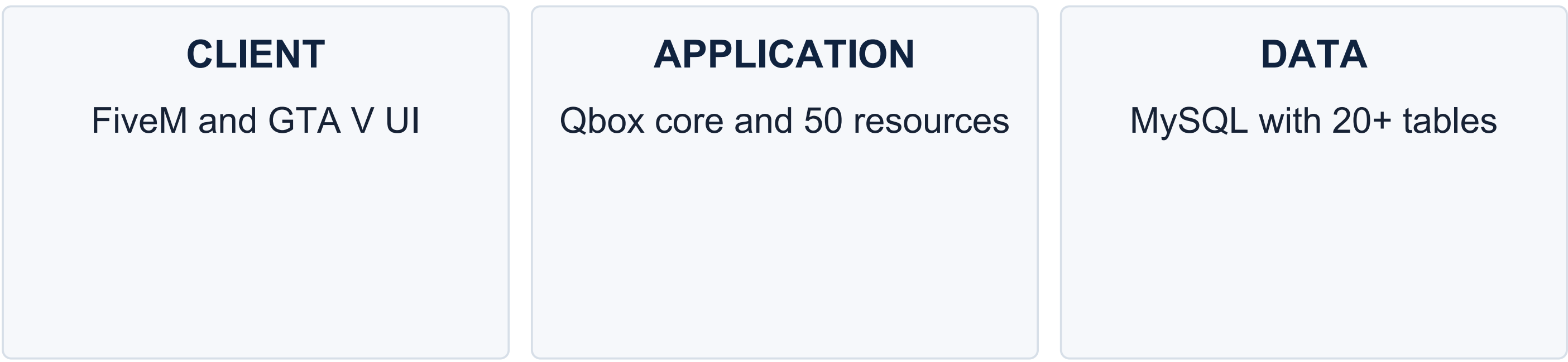


txAdmin resource monitor and server control

## APPLICATION WORKFLOW



## SYSTEM ARCHITECTURE



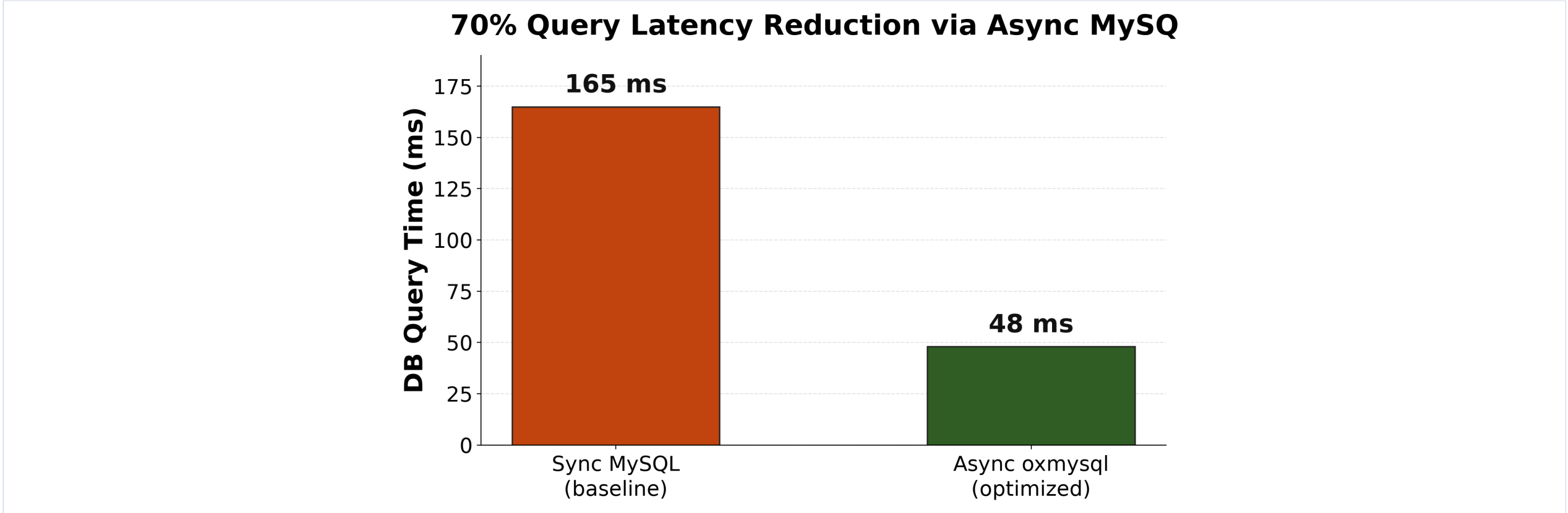
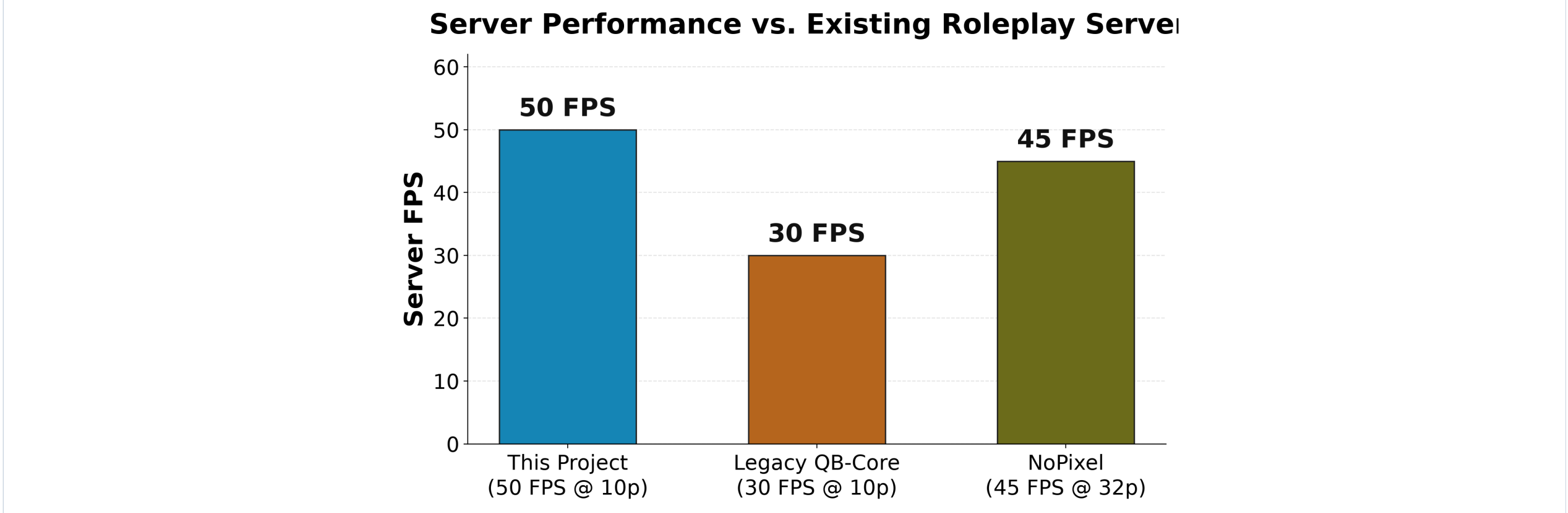
## RESULTS AND DISCUSSION

**50-60  
SERVER FPS**

**0  
SUCCESSFUL EXPLOITS**

**8.2/10  
USER SCORE**

A stable 10-player deployment (scalable to 32) maintained 99% uptime. All 13 job systems passed functional tests with no money duplication.



## CONCLUSION AND FUTURE WORK

The server provides an integrated, secure, and scalable roleplay environment that outperforms the team's legacy baseline in performance and database response. Future work includes an interactive tutorial, additional jobs, player-owned businesses, and multi-server clustering for more than 32 players.

## ACKNOWLEDGEMENT

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Bangladeshi Banking



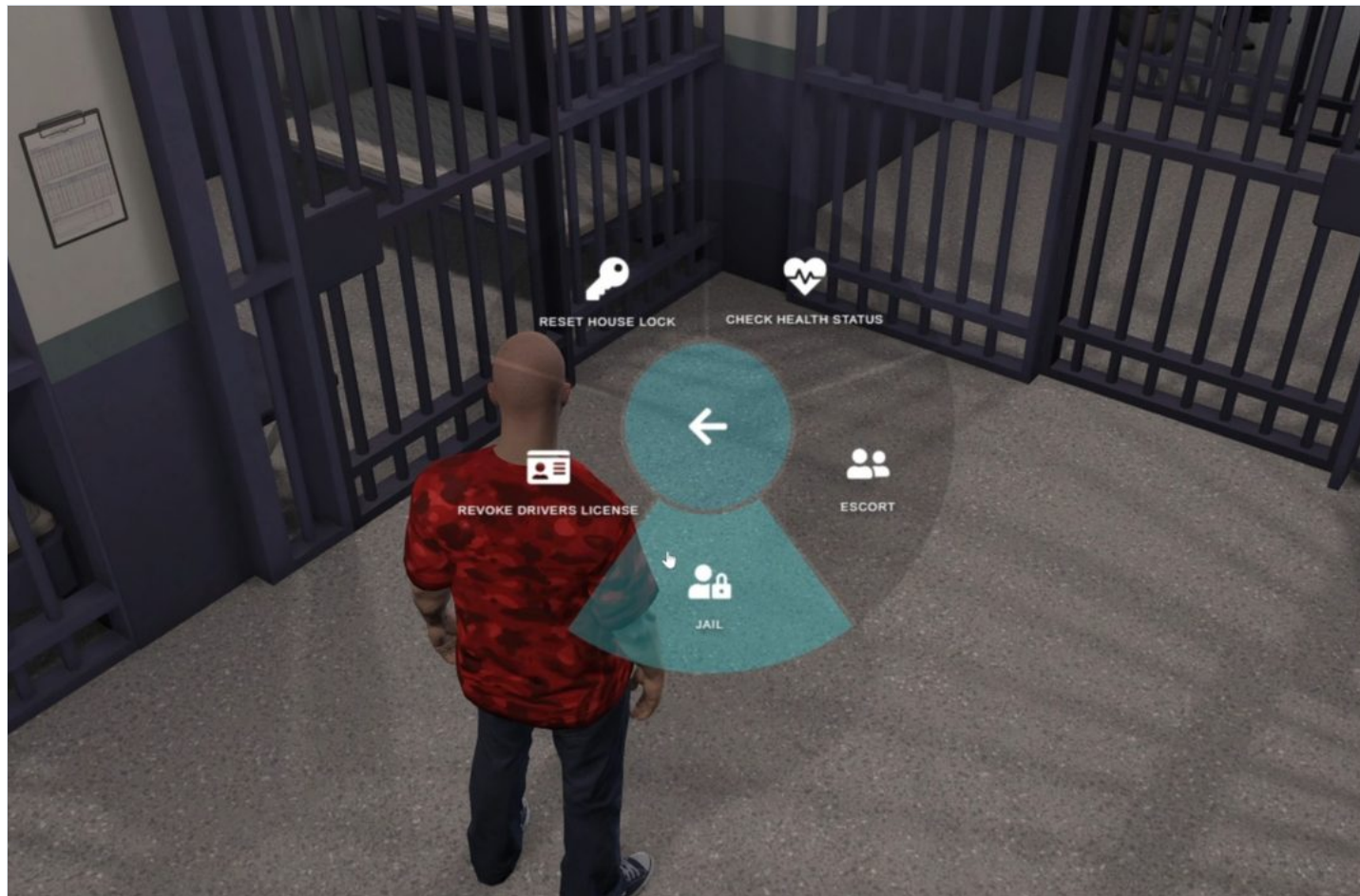
NPWD Phone



Emergency Department



Vehicle and Garage



3D Voice and Radio



Localized City Branding